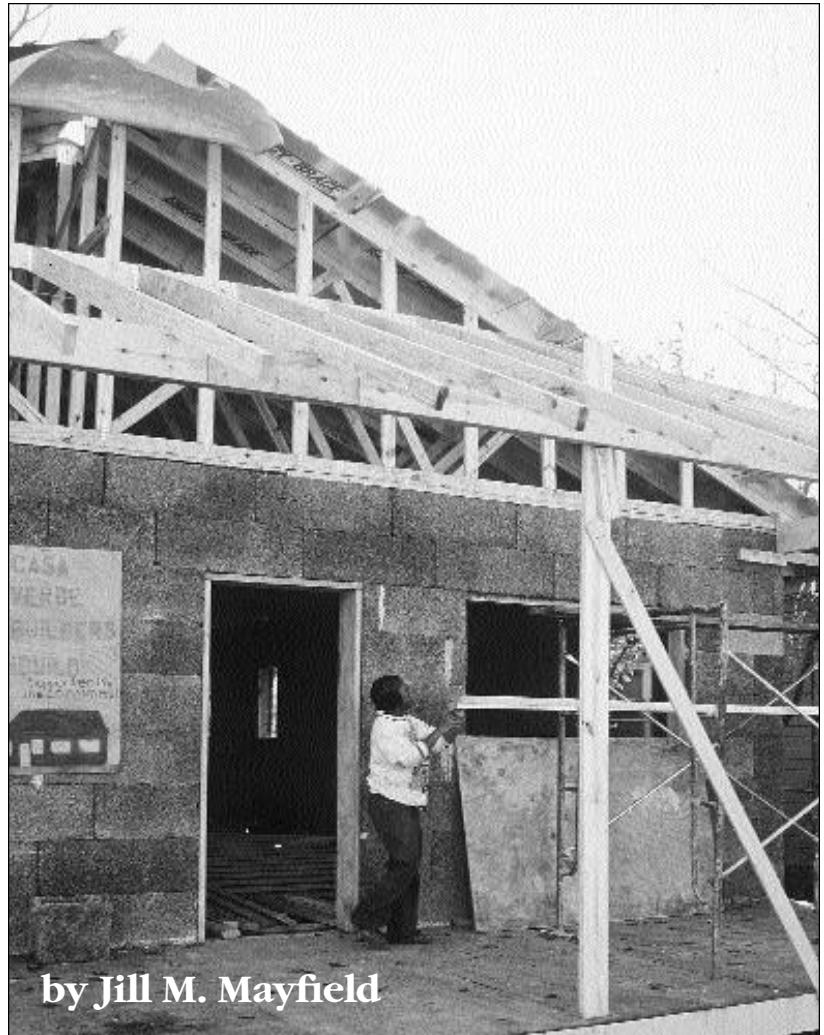


Green Building Programs—A Growing Trend

Resource- and energy-efficient building programs are taking seed across the nation and drawing in more traditional builders.



In 1991, Austin, Texas, and the city's municipally owned electric utility, Austin Energy, launched the nation's first green building program, initiating a trend that has slowly spread north, east, and west across the country. At that time green building—using construction practices and materials designed to limit a building's use of environmental resources—wasn't even a blip on the radar screen of the nation's largest residential building trade organization, the National Association of Homebuilders (NAHB). Nine years later, green building has almost hit the big time. Seven well established green building programs are thriving across the country, a new program is taking off in Atlanta, and many other communities are beginning such programs (see "Atlanta's Green Building Program

Breaks Ground," p. 36). NAHB hosted the first national Green Building conference in Denver, Colorado, last year. All the signs suggest that more sustainable building methods are more than just a passing fad.

"Although I know of no official statistics to back up the claim, I use the number of local builder programs and the inquiries I receive here at the research center as indicators that the field is growing," says Peter Yost, director of resource and environmental analysis at the National Association of Homebuilders Research Center in Washington, D.C. Yost says he gets about three green building-related calls each week from builders, home buyers, manufacturers, and home builders associations. "The field is growing because both builders and buyers are becoming educated regarding the value of resource-

efficient home design, construction, and operation," says Yost.

In April 1999, in response to an increasing number of requests, the NAHB Research Center produced *Guide to Developing Green Building Programs*. This 120-page guide describes program development and green building techniques. It discusses costs, technologies, availability, and practicality; and it includes references for everything from framing materials to xeriscaping. More than 150 copies of the guide have been distributed to date.

Across the country, more than 6,000 homes have been built that qualify for one of the nation's green building programs (see Table 1). It is impossible to state the total amount of energy and resources saved through these programs, because most of the programs don't calculate such savings. However,

each of these homes cuts back on the resources needed to build or operate a house and also provides a model for the community of builders and prospective homeowners in the area. And while most of the programs don't yet take into consideration such sustainable community concerns as reducing travel by vehicles, some programs do address these issues. For example, the Build a Better Clark Program in Washington has created a program area for developers, with special consideration given to pedestrian friendliness and site ecosystem protection. And the city of Austin's green building program gives additional credit for such sustainable community features as public transit within a 15-minute walk, location of a subdivision near a hike-and-bike trail or park, or the presence of a backyard composting bin.

Marketing Change

In trying to motivate builders to rethink traditional practices in new residential construction, green building programs have chosen to abandon the traditional means of causing change in building practices: the adoption of codes and ordinances that force compliance with a local standard. This approach can be successful, but it frequently creates clashes between the governing body and the regulated party that can delay or compromise the enactment of environmental and conservation initiatives.

"Codes are for life and safety issues," says Yost. "The best way to achieve more



Rainwater harvesting is a feature that is frequently offered by builders who participate in green building programs.

resource-efficient construction is to demonstrate the value to builders and buyers and have the market drive this phenomenon." Developers of green building programs have tended to agree with Yost's approach, creating voluntary marketing programs that entice builders to build homes that are energy-, water-, and resource-efficient, and environmentally sound.

Today, green building programs fall into two categories. Six out of the eight programs now in existence are extensions of the local, state, or regional homebuilders association. These programs have arisen for reasons as diverse

as the communities themselves, but all seek to bring marketing attention and positive public opinion to area builders. Many have also been motivated by the desire to head off accusations by local environmental groups that builders don't care about the environment.

The remaining two green building programs are affiliated with a city or county government. These programs have been developed by government officials who want to draw attention to the benefits of green building for both the local citizens and the local environment. To some extent, they were also developed to deal with problems municipalities have faced in passing development and environmental regulations.

Austin Takes an Early Lead

The program that started it all was the City of Austin Green Builder Program, now known as the Austin Energy Green Building Program, in Austin, Texas. The program is a division of Austin Energy, the city's municipally owned electric company. It evolved out of the city's aggressive energy conservation programs, which stressed demand-side management. The Austin program has rated 1,381 homes since 1994, when the most current version of the rating system was started. On average, each home rated saves an estimated 1.33 kW.

"Green building is moving into the mainstream, where it is starting to transform the market," says Marc Richmond-Powers, project manager for the Austin program. "Buyers of homes and com-

Table 1: Active Voluntary Green Building Programs in the United States

Location	Program Name	Administration	Year Started	Number of Homes in Program
Austin, Texas	Austin Energy Green Building Program	Government	1991	1,381
Scottsdale, Arizona	Sustainable Building Program in the Sonoran Desert	Government	1998	25
Colorado	Built Green Colorado	Home builders association	1995	4,000
Kitsap County, Washington	Build a Better Kitsap	Home builders association	1997	50
Clark County, Washington	Build a Better Clark	Home builders association	1998	11 homes, 2 developments
Central New Mexico	Green Builder Program	Home builders association	1997	800
Southern Maryland	Builder Green	Home builders association	1999	NA
Atlanta, Georgia	Earth Craft House	Home builders association	1999	NA

Source: Mary James/National Association of Homebuilders Research Center

mercial buildings are now asking for green features and buildings when they come to an architect's office."

The Austin program has two categories of membership: members, which includes builders, architects, and developers; and associate members, which includes material suppliers, real estate agents, lenders, and service providers. Currently no fees are charged for membership or to rate individual homes. Members must attend an initial orientation and then attend at least two of the free monthly technical seminars held each year to maintain membership. Members must also agree to promote green building to their clients and to participate actively by having their homes rated by the program. Currently, just over 100 builders and architects are members of the Austin program.

The Austin program rates buildings in four areas: residential, multifamily, municipal, and commercial. The residential program rates new and remodeled homes for energy efficiency, water efficiency, resource efficiency, health and safety, and community. Ratings range from one to five stars, with five being the most green. All projects must meet a baseline in each of the rating areas. Once above the



This 1,600 ft² home exceeded the highest rating in the Build a Better Kitsap Program.

baseline, builders may choose any of various weighted options. The more options chosen, the higher the final rating. Staff members periodically verify ratings by visiting sites and by reviewing construction documents. The Sustainable Building Sourcebook is the Austin program's foundation document. This reference, which runs to more than 300 pages, describes in

detail the building practices promoted by the program.

Marketing is a key component of the Austin program, as it is with all voluntary green building programs. Members are provided signs and window stickers for each rated home and a rating certificate for the buyer. Members of the program are promoted by the city through radio, print, and television advertising. Special events, such as a tour of green homes, provide a way for builders to showcase their homes and for the program to generate awareness in the community.

In the commercial, municipal, and multifamily program areas, staff members offer free consulting services to designers and planners to help implement sustainable features in their buildings and developments. Although there is no rating system for commercial, municipal, or multifamily projects, they must follow a set of guidelines in order to be considered green.

Colorado Builds Green

The Built Green Colorado program began in 1995 as the Green Builder Program of Metro Denver. It was created through the joint efforts of the Home Builders Association (HBA) of

Atlanta's Green Building Program Breaks Ground

The Greater Atlanta Homebuilders Association's Earth Craft House is the latest green building program to take shape. Introduced in November 1999, the Earth Craft House is a builder training and education program to encourage environmentally friendly building techniques. Builders who sign up for the program attend a one-day training course provided by the program's partner, the Southface Energy Institute. Each registered builder must meet certain criteria, which include a minimum number of options set forth in the checklist of construction and material alternatives. In return, the HBA of Atlanta will provide marketing support and materials to members who are participating in the program.

For a green building program that is just getting off the ground, the Earth Craft House is already exceeding expecta-

tations. "We were hoping to have 50 Earth Craft Houses built in the first year," says Dennis Creech, executive director of the Southface Energy Institute. "We already have builder commitments to put up 500 Earth Craft Houses." Creech attributes the program's early success to a well-developed marketing campaign and to the fact that the program's design makes it accessible to mainstream builders. "This program was not intended to push the fringe," says Creech. The program's training workshop shows builders how they can incorporate green building techniques into their building practices and still be profitable.

Another reason that builders have accepted the Earth Craft House program so readily is that builders participated in its design. In particular, builder input strongly influenced the certification process, according to Creech. "The

builders wanted to make sure that if they were building to meet Earth Craft standards, everyone else in the program was also adhering to those standards," he says. For every Earth Craft House, the builder and the home buyer must sign off on a worksheet that details how the house meets the program's standards. The documentation used to support the certification process, such as load calculations and warranties, will be part of a home buyer's closing documents.

For at least the first year of the program, Southface raters will conduct a final inspection that includes a blower door and Duct Blaster test. Earth Craft House sponsors—the Georgia Environmental Facilities Authority, Home Depot, Andersen Windows, and Georgia-Pacific—will underwrite the cost of these inspections for the first year of the program.

Metro Denver, the Boulder chapter of the HBA, the governor's Office of Energy Conservation (OEC), the E-Star Home Energy Rating Program (E-Star), and Public Service Company of Colorado (PSCo).

Built Green Colorado is a voluntary program that uses buyer demand and market education to motivate builders to produce homes that are more energy-efficient, more durable, and more comfortable, and that have better indoor air quality than the average home. Builders are also encouraged to take measures that conserve water, reduce pollution, and save natural resources in the construction and operation of the homes.

In the Built Green program, the builder registers individual homes to receive the Built Green designation. Each registered home must receive an energy rating of at least four stars as certified by Energy Rated Homes of Colorado or be built to the Model Energy Code 1993 (MEC 93) standards. To qualify as Built Green, each registered home also must include a minimum number of the green options in the Built Green checklist. The checklist has 138 separate features in 21 categories. A builder must choose at least 35 features if all categories are used, or 38 features if one or more categories are skipped. The Colorado program has registered more qualified homes than any other program—roughly 4,000.

Doug Seiter, state coordinator for the program, prefers this point system to a multitiered rating system. "It's a challenge to make a multitiered system attractive to everyone. The Built Green program point system is more inclusive and allows the builder much more flexibility," says Seiter. That flexibility and the support of the Denver HBA have been key to the Colorado program's active builder—even production builder—participation, according to Seiter. "You have to have a program that is attainable at a production level," says Seiter. Colorado's program has faced criticism that the criteria are too easy. "My answer to that criticism is you're not going to have a big impact on the building industry if the criteria are too stringent." Seiter is quick to add that a program wouldn't be



Austin Energy Green Builder Greg Mangum is constructing this home using Greenblock Monolithic Modular Units. A type of insulated concrete form, Greenblock is made of flame retardant expanded polystyrene that has been expanded by steam. It has no ozone depleting chemicals.

valuable if its criteria were limited to standard construction practices. "It's a balancing act," he says.

With the participation of larger builders, Seiter sees the program's effects trickling down through the industry, even to suppliers. "The whole building industry is changing, and building products as well," says Seiter. He adds that recent media attention given to green building has made today's consumer more aware and more likely to make good choices when purchasing a home. "Given the choice, who wouldn't go with the better built home?" says Seiter.

Sustainable Building in the Sonoran Desert

In Scottsdale, Arizona, home builders and prospective home buyers interested in environmentally compatible homes can turn to the city's Green Building program, which is also known as the Sustainable Building Program in the Sonoran Desert to emphasize the unique sensitivity of the desert environment. Scottsdale's program rates homes in six environmental impact areas: site use, building materials, solid waste, energy, indoor air quality, and water.

A green building checklist and point system is used to qualify homes. The checklist contains 150 green building options. Like Built Green Colorado, Scottsdale's is an all or nothing program. A project is considered green if it earns at least 50 points on the checklist.

A builder or developer may enter any number of homes in any given subdivision into the program. The program is voluntary and open to all home builders in Scottsdale. There is discussion among public and private agencies about expanding the program into other communities around the valley and around the state.

Every builder and designer who enters a home into the program is required to attend a city-sponsored green building workshop or seminar. The workshop serves as an introduction to energy- and resource-efficient and environmentally responsible buildings. It features building experts in such areas as site use, energy, building materials, indoor air quality, water, and solid waste.

As a consumer-driven program, Scottsdale engages in an ongoing marketing effort to bring the program to the attention of the home buying public. For Scottsdale's green builders,

the program provides promotional packages, expedited plan review, job site signs, a directory of local materials and products that meet program criteria, workshops and seminars, and information on how to obtain energy-efficient mortgages and a solar-energy tax credit of 25% of the cost of an installed solar energy system up to a maximum of \$1,000.

Washington's Mushrooming Green Building Programs

Build a Better Kitsap is a program of the Home Builders Association of Kitsap County, Washington, that promotes the use of environmentally friendly features in new and remodeled homes. Like Austin, Kitsap assigns a rating to homes. Homes enrolled in the program are rated by the association based on the number of environmental features included in the home. Construction projects that meet program standards

are awarded a one-, two-, or three-star rating certificate, with three stars representing the highest level of achievement. The energy-efficient features used in a 2,400 ft² home that recently qualified for the three-star rating are expected to slash the heating bill by 50%. Without these features, this home would typically cost \$1,500 per year to heat, but computer modeling shows that with them, the new homeowner will pay only \$750 per year.

In a unique twist, the Build a Better Kitsap program provides builders with an environmentally responsible homeowner's manual to be handed to the buyers of qualifying homes. The builder can customize this manual, which provides tips on such practices as composting and recycling, with information on the green features of an individual home.

While the Kitsap program has been successful, it has remained small, because the area has a stable population, and growth in the housing industry there has been very slow. Clark County,

Washington, which is experiencing more rapid growth, has recently begun a green building program using the same guidelines and materials as Kitsap. According to officials in the Kitsap HBA, the Build a Better Clark program is catching on quickly. Seattle is also catching on. The Master Builders Association of King and Snohomish Counties is putting together a similar green building program, which is expected to be functioning by late spring.

In the less than ten years since it began, green building has moved in from the fringe and is now approaching the mainstream. Builders and building officials around the country are asking questions about green building—and asking how they can incorporate efficient, durable, and environmentally sound construction into their communities. 

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